

# The Two-Site Inclusion Process: Complete Hahn Spectrum and the Absorbing Face

HCS-C326 theorem-and-evidence package

3 September 2026 — revision round 1

## Abstract

We diagonalize the convention-locked two-site symmetric inclusion process. The result includes its beta-binomial law, every Hahn mode and finite-sum norm, the full semigroup kernel, sharp  $L^2$  decay, and the singular zero-parameter absorbing limit. Exact rational grids audit, but do not prove, the theorem.

## 1 Frozen generator and reversible law

Fix  $N \in \mathbb{Z}_{\geq 0}$  and  $\alpha > 0$ . State  $x \in S_N = \{0, \dots, N\}$  is the first-site occupancy; the second is  $N - x$ . Our unscaled continuous-time generator is

$$(Lf)(x) = b_x[f(x+1) - f(x)] + d_x[f(x-1) - f(x)], \quad b_x = (N-x)(\alpha+x), \quad d_x = x(\alpha+N-x), \quad (1)$$

with unavailable endpoint terms zero. Put

$$\pi(x) = \frac{N!(\alpha)_x(\alpha)_{N-x}}{x!(N-x)!(2\alpha)_N}. \quad (2)$$

**Theorem 1.** *For  $N \geq 1$ , (2) is the unique reversible law. Define*

$$H_j(x) = {}_3F_2\left(\begin{matrix} -j, j+2\alpha-1, -x \\ \alpha, -N \end{matrix}; 1\right), \quad h_j = \sum_{x=0}^N \pi(x) H_j(x)^2, \quad 0 \leq j \leq N.$$

*Then  $h_j > 0$ , the  $H_j$  are a complete orthogonal basis, and*

$$LH_j = -\lambda_j H_j, \quad \lambda_j = j(j-1+2\alpha).$$

*The complete transition kernel is*

$$p_t(x, y) = \pi(y) \sum_{j=0}^N e^{-\lambda_j t} \frac{H_j(x) H_j(y)}{h_j}. \quad (3)$$

*For centered  $f$ ,  $\|e^{tL}f\|_{2,\pi} \leq e^{-2\alpha t} \|f\|_{2,\pi}$ ; the exponent is sharp. At  $\alpha = 0$ , the endpoints absorb, absorption is almost sure, and  $\mathbb{P}_x(X_\infty = N) = x/N$  for  $N \geq 1$ . Moreover  $\{c\delta_0 + (1-c)\delta_N : 0 \leq c \leq 1\}$  is the complete stationary family, and  $\pi_{\alpha,N} \Rightarrow (\delta_0 + \delta_N)/2$  as  $\alpha \downarrow 0$ . For  $N = 0$  the process and law are the singleton.*

The adjacent identity  $\pi(x)b_x = \pi(x+1)d_{x+1}$  follows by cancelling Pochhammer factors; Chu–Vandermonde gives the denominator in (2). For  $N \geq 1$  all interior edge rates are positive, proving irreducibility, uniqueness, and self-adjointness in  $L^2(\pi)$ .

The evidence grid uses  $\alpha \in \{1/2, 1, 3/2, 2\}$  and  $0 \leq N \leq 8$ : 36 parameter rows, 180 states, and 180 full Hahn vectors. All entries are exact rationals independently reconstructed; finite evidence is not a proof.

## 2 Full Hahn diagonalization

**Lemma 1.** *The degree- $j$  polynomial filtration is invariant under  $L$ , and the diagonal entry on its one-dimensional quotient is  $-j(j-1+2\alpha)$ . The terminating series defining  $H_j$  has degree  $j$ , satisfies  $LH_j = -j(j-1+2\alpha)H_j$ , and obeys  $H_j(0) = 1$ .*

*Proof.* For  $f(x) = x^j$ , the prospective  $x^{j+1}$  terms in (1) cancel. The  $x^j$  coefficient is

$$j[(N-\alpha) - (N+\alpha)] - j(j-1) = -j(j-1+2\alpha).$$

For identification, write the finite expression

$$H_j(x) = \sum_{k=0}^j \frac{(-j)_k (j + 2\alpha - 1)_k (-x)_k}{(\alpha)_k (-N)_k k!}.$$

Put  $\phi_k(x) = (-x)_k$ . Direct forward and backward differencing gives

$$L\phi_k = -k(k - 1 + 2\alpha)\phi_k - k(N - k + 1)(\alpha + k - 1)\phi_{k-1}.$$

For  $H_j = \sum_k c_k \phi_k$ , the eigenvalue equation requires

$$\frac{c_{k+1}}{c_k} = \frac{\lambda_j - \lambda_k}{(k + 1)(N - k)(\alpha + k)} = \frac{(j - k)(j + k + 2\alpha - 1)}{(k + 1)(N - k)(\alpha + k)}.$$

This is exactly the ratio of consecutive coefficients in the displayed terminating sum, by  $(a)_{k+1} = (a + k)(a)_k$ . Consequently every coefficient on the left of

$$b_x[H_j(x + 1) - H_j(x)] + d_x[H_j(x - 1) - H_j(x)] + j(j - 1 + 2\alpha)H_j(x)$$

is zero. This proves the finite difference identity. Its denominators do not vanish for  $j \leq N$  and  $\alpha > 0$ ; its final coefficient is nonzero, while substitution  $x = 0$  gives one.  $\square$

Because  $\lambda_{j+1} - \lambda_j = 2(j + \alpha) > 0$ , these are all the distinct diagonal values from Lemma 1. Self-adjointness makes their eigenvectors orthogonal. There are  $N + 1$  of them, hence they are complete, and each displayed finite-sum norm  $h_j$  is positive. Expanding any function in this normalized basis proves (3). Parseval on the constants' orthogonal complement gives the stated bound because  $\lambda_1 = 2\alpha$ ;  $H_1(x) = 1 - 2x/N$  gives equality and therefore sharpness. When  $N = 1$ , this says exactly that the spectrum is  $\{0, 2\alpha\}$  and  $\pi = (1/2, 1/2)$ .

## Revision certificate: complete Hahn spectrum and semigroup

This round adds polynomial triangularity, the terminating difference identity, orthogonality with positive finite-sum norms, the full kernel, and sharp decay.

## References

- [1] C. Giardinà, F. Redig, and K. Vafayi, Correlation inequalities for interacting particle systems with duality, *J. Stat. Phys.* 141 (2010); DOI: 10.1007/s10955-010-0055-0; arXiv:0906.4664.
- [2] NIST Digital Library of Mathematical Functions, Sections 18.19, 18.20.5, and 18.22(ii), Hahn definitions, representation, and difference equations.